Pyramakerz®

Unit 1

Lesson 4

Spark The Challenge!

Unlocking Ultrasonic Secrets.

Time Frame:[90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Differentiate between sound waves.
- Explain how ultrasonic sensors use sound waves to detect objects.
- Describe the concept of echolocation and how it relates to ultrasonic sensors.
- Use estimation to calculate distance using the speed of sound and time measurements.
- Assemble a basic electronic circuit that includes an Arduino, ultrasonic sensor, and buzzer.
- Write and modify Arduino code that triggers a buzzer response based on measured distance.

Challenge questions of the lesson.

1. **How do sound waves travel, and what makes ultrasonic waves different from regular sounds we hear every day?**
2. **If an animal like a bat can "see" using sound, how might a sensor do something similar to detect nearby objects?**
3. **How could you estimate the distance of an object using only sound and time? What information would you**

SEL Relation

1. Self-Awareness
   - **Recognizing Emotions:** As students encounter challenges—like a buzzer that won't sound—they learn to notice feelings of frustration or excitement.
   - **Growth Mindset:** Emphasize that debugging code and circuits is part of learning. Encourage students to view mistakes as opportunities to grow.
2. Self-Management
   - **Perseverance:** Building circuits and writing code requires patience. Students practice sticking with a problem until they find a solution.
   - **Goal-Setting:** Each objective (e.g., "Make the buzzer beep when an object is within 20 cm") gives a clear target, helping students plan steps and manage their time.
3. Social Awareness
   - **Empathy for Users:** By thinking about how ultrasonic "echolocation" aids people with vision loss, students develop compassion for others' experiences.
   - **Perspective-Taking:** Discuss real-world applications (like smart glasses) to understand how technology can meet diverse needs..
4. Responsible Decision-Making
   - **Ethical Use of Technology:** Reflect on how their inventions could improve independence and safety for people with disabilities.
   - **Reflective Practice:** After testing, students evaluate what worked, what didn't, and plan improvements, strengthening critical thinking.

Engage

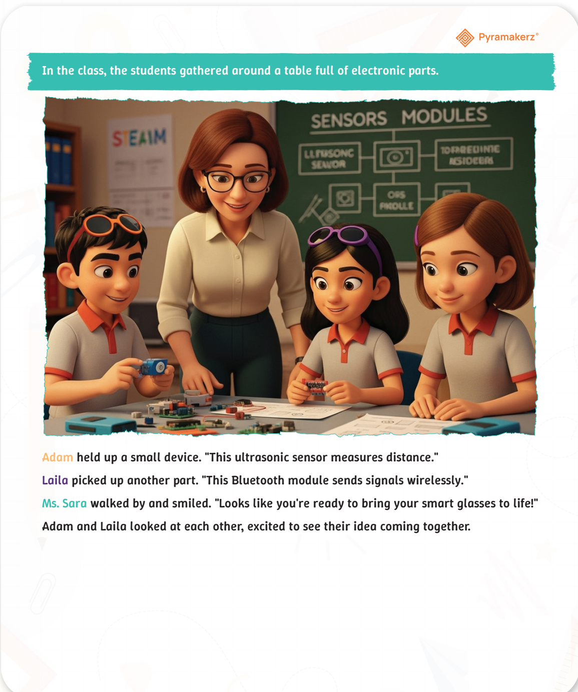
Teacher Role:

- **Set the Scene:** Read the Scene 1 dialogue aloud with enthusiasm, using distinct voices for Adam, Laila, and Mrs. Sara to draw students into the story of building smart glasses.
- **Activate Curiosity:** Point to the “Let’s Think” prompt and invite students to consider the big question: “Why can we hear some sounds but not others?”
- **Guide Discussion:** Ask probing follow-up questions—“What do you think makes a sound too high or too low?”—and encourage students to connect this to the ultrasonic sensor in the story.
- **Bridge to Next Activity:** Explain that understanding sound waves will help them program their own sound-alert system with the ultrasonic sensor and buzzer.

Expected From Students:

- **Active Listening:** Pay attention to the dialogue and identify the key components—ultrasonic sensor and Bluetooth module—needed to build smart glasses.
- **Thoughtful Reflection:** Respond in writing to the “Let’s Think” question, describing factors (like pitch and frequency) that affect what we hear.
- **Participate in Discussion:** Share ideas about why human ears can’t hear ultrasonic frequencies and how sensors can “hear” beyond our range.
- **Make Connections:** Link their observations about sound to the upcoming hands-on circuit and coding activities.

Parts of the books included.



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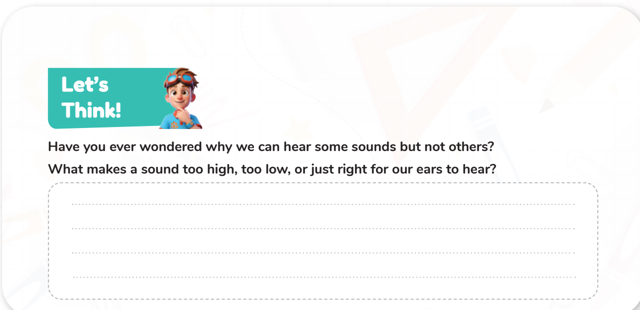
In the class, the students gathered around a table full of electronic parts.

Adam held up a small device. "This ultrasonic sensor measures distance."

Laila picked up another part. "This Bluetooth module sends signals wirelessly."

Ms. Sara walked by and smiled. "Looks like you're ready to bring your smart glasses to life!"

Adam and Laila looked at each other, excited to see their idea coming together.



Let's Think!

Have you ever wondered why we can hear some sounds but not others?

What makes a sound too high, too low, or just right for our ears to hear?

Story

1. **Warm Welcome & Quick Recap.**
   - "Good morning, innovators! Yesterday we dreamed up smart glasses to help people who can't see clearly. Who remembers two key parts we talked about?"
   - (Listen for "ultrasonic sensor" and "Bluetooth module," and jot them on the board.)
2. **Introduce Today's Scene with Expression.**
   - "Today, Adam and Laila are back in the lab—surrounded by real electronic parts! Let's see what they discover."
   - (Read the Scene 1 dialogue aloud, using a bright, curious tone for Adam and Laila, and a supportive voice for Mrs. Sara.)
3. **Bridge to Inquiry.**
   - "Adam says, 'This ultrasonic sensor measures distance.' Laila adds, 'This Bluetooth module sends signals wirelessly.' What question do you think they'll ask next?"
   - (Prompt: "How does sound help us 'see' obstacles?")

Possible Strategies for Implementation:

1. **Think–Pair–Share**
   - Think: Students take 30 seconds to jot down what they know about sensors and sound.
   - Pair: Turn to a neighbor to compare ideas.
   - Share: Invite a few pairs to share out, reinforcing correct terminology.
2. **"Sound Detective" Demo**
   - Play a short audio clip of a high-pitch tone and a low-pitch tone.
   - Ask: "Which could an ultrasonic sensor hear that we can't?"
   - Relate back to how bats or dolphins use echolocation—and how our sensor works similarly.
3. **Storyboard Sketch**
   - On mini whiteboards or sticky notes, students draw a quick sketch of a sensor "listening" for an object and sending data via Bluetooth.
   - Hang their sketches on a "Design Wall" to revisit during the Build phase.
4. **KWL Chart Update**
   - In the "K" column (What I Know), add "Ultrasonic sensors measure distance using sound."
   - In the "W" column (What I Want to Know), students write: "How does the sensor turn sound into a number we can read?"
   - Use this to guide the next hands-on activity.

Let's Think!

Objectives:

Students will explore the concept of sound frequency and pitch by reflecting on why some sounds are inaudible to human ears, laying the groundwork for understanding ultrasonic sensing.

Possible Answers

- "Sounds are made of vibrations. If vibrations happen too fast (high frequency), our ears can't catch them."
- "If vibrations are too slow (low frequency), they're below what our ears can detect."
- "Our ears hear best in a certain middle range—too high or too low and it becomes silent to us."
- "Animals like bats use very high frequencies (ultrasound) to "see" with sound."

Activity Instructions:

1. Read the Prompt:

- Display the "Let's Think" question:
  - "What makes a sound too high, too low, or just right for our ears to hear?"

2. Silent Reflection:

- Give students 1–2 minutes to write their initial thoughts in the space provided.

3. Pair Discussion:

- Have students turn to a partner and share their ideas, adding any new insights.

4. Whole-Class Share:

- Invite a few volunteers to read their responses aloud. Record key terms (vibration, frequency, pitch) on the board.

Possible Strategies for Implementation:

- Think–Pair–Share: Structure reflection and discussion so every student has a chance to formulate an idea before sharing.
- Analogies & Demonstrations: Use a guitar or tuning fork to show high and low pitches, letting students hear the difference firsthand.
- Visual Frequency Chart: Draw a simple graph on the board showing low to high frequency and label the audible range for humans (~20 Hz to 20 kHz).
- Animal Connection: Show a quick image or video of a bat echolocating to anchor the idea of ultrasonic sound beyond human hearing.

Explorer's Toolkit

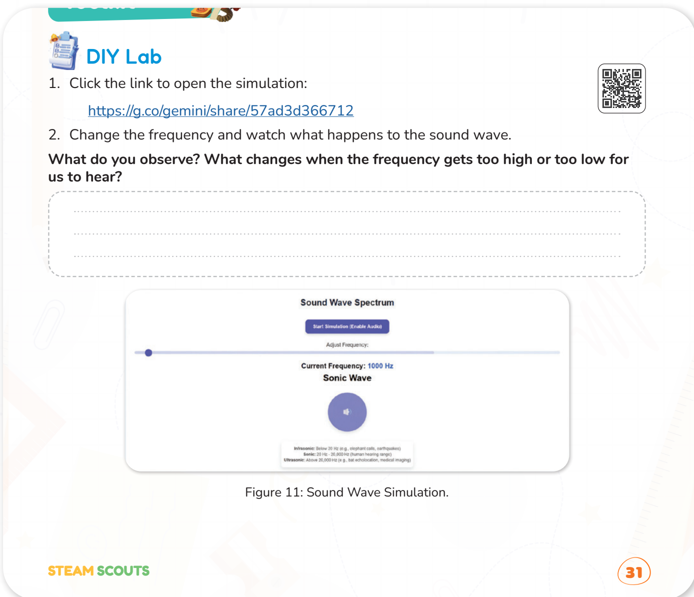
Teacher Role:

- **Set Up & Demonstrate:** Project or distribute the simulation link and walk students through how to adjust the frequency slider.
- **Facilitate Observation:** Ask guiding questions as students work: "What do you notice about the wave shape?" "How does the sound change when you move the slider?"
- **Support Reflection:** Circulate and prompt students who are stuck—"How does this compare to the sounds you hear every day?"
- **Connect Back to Story:** Remind students that understanding these invisible waves helps make ultrasonic sensors work, just like the ones in Adam and Laila's smart glasses.

Expected From Students:

- Click and explore the online sound wave simulator.
- Change the frequency across infrasonic, sonic, and ultrasonic ranges.
- Record observations about wave behavior and audibility when frequencies exceed human hearing limits.
- Use their observations to explain how an ultrasonic sensor can "hear" distances we cannot.

Parts of the books included.



DIY Lab

1. Click the link to open the simulation:

   https://g.co/gemini/share/57ad3d366712

2. Change the frequency and watch what happens to the sound wave.

What do you observe? What changes when the frequency gets too high or too low for us to hear?

Figure 11: Sound Wave Simulation.

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DIY Lab

Objective:

Students will investigate how sound wave frequency affects our ability to hear and relate those changes to how ultrasonic sensors detect objects beyond human hearing range.

Activity Instructions

1. Open the Simulation

- Go to: https://g.co/gemini/share/57ad3d366712

2. Adjust Frequency

- Move the slider slowly from low (infrasonic) through sonic into ultrasonic range.

3. Observe & Record

- Note the wave's appearance (wavelength and amplitude).
- Write responses to:
  - "What happens when the frequency is too low (infrasonic)?"
  - "What happens when the frequency is too high (ultrasonic)?"

4. Explain the Connection

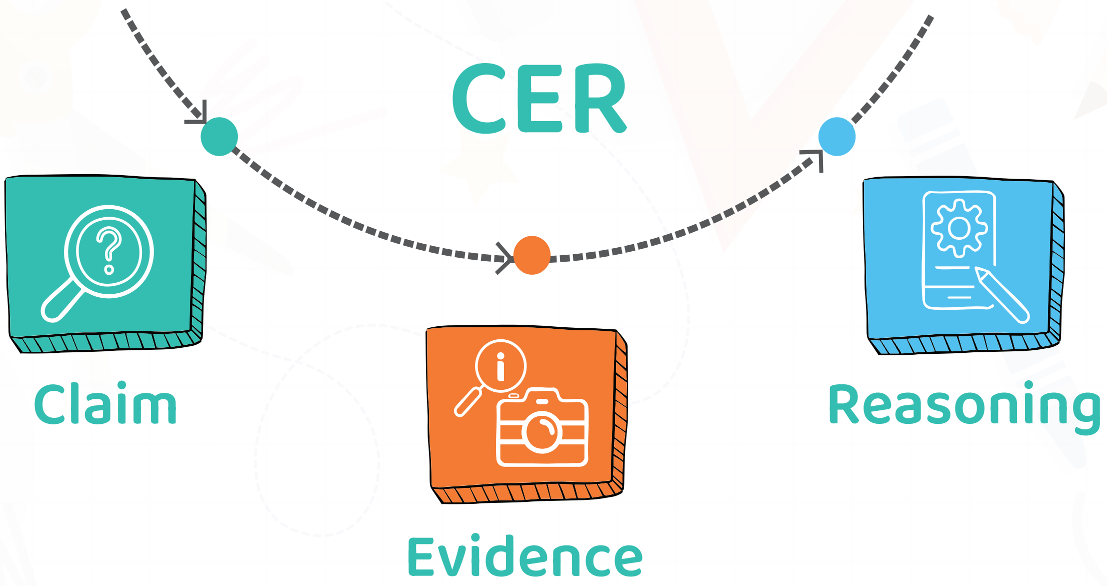
- In your notebook, describe why humans can't hear ultrasonic sounds and how sensors use them to measure distance.

Possible Strategies for Implementation:

- **Think–Pair–Share:** Students first jot down observations individually, then compare notes with a partner before sharing with the class.
- **Guided Inquiry:** Provide a table with columns for "Frequency," "Wave Shape," "Audible?" and "Notes" to scaffold data collection.
- **Physical Analogy:** Use a slinky to model low vs. high-frequency waves in class before simulation.
- **Exit Ticket:** Ask students to write one sentence explaining how ultrasonic sensors "listen" at frequencies we cannot hear.

Possible Answer:

- As I slide the frequency from low to high, I see the wave peaks get closer together—meaning the wavelength is shortening.
- In the very low (infrasonic) range, the tone disappears and I can't hear anything; the wave pattern is very "stretched out."
- In the mid-range (around 20 Hz–20 kHz), I clearly hear a tone that rises in pitch as the frequency increases; the waves wiggle at a speed my ears can detect.
- Once I go above about 20 kHz (ultrasonic), the wave pattern is extremely tight and fast, and the tone vanishes again—my ears can't pick up sounds that high.



What do you think is happening?

When we change the frequency slider, the wave's wavelength and our ability to hear it also change—very low and very high frequencies become inaudible, while middle-range frequencies produce a clear tone.

What did you see in the simulation?

- At 1 Hz (infrasonic range), the wave peaks were spaced very far apart and I could not hear any sound.
- Around 1,000 Hz (within the human hearing range), the wave crests moved closer together and I heard a distinct pitch that rose as I increased frequency.
- Above about 20,000 Hz (ultrasonic range), the wave became extremely tightly packed again and the tone disappeared—no sound was audible.

Why does this happen?

Human ears can only detect frequencies roughly between 20 Hz and 20 kHz. When the simulation's frequency falls below that range, the wave is too slow (infrasonic) for our ears to pick up. When it goes above that range, the wave is too fast (ultrasonic) for our ears to process. Thus, only waves whose frequency lies in the middle range produce an audible tone, matching what we experienced in the simulation.

Explain

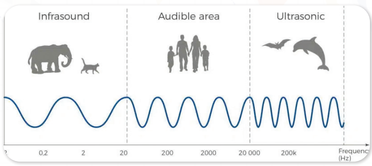
Teacher Role:

- Guide students through exploring how sound frequency affects hearing and perception.
- Pause the YouTube video at key moments to ask clarifying questions or allow time for note-taking.
- Facilitate discussion on infrasound, audible sound, and ultrasound.
- Prompt connections between animal echolocation and ultrasonic sensors.
- Encourage curiosity by introducing the National Geographic article and helping students extract key facts.

Expected From Students:

- Actively watch the video and note differences between infrasound, audible sound, and ultrasound.
- Observe how humans and animals perceive different sound frequencies.
- Read the National Geographic article and identify at least one fact about echolocation in animals.
- Discuss how animals like bats or dolphins use sound waves in ways similar to how ultrasonic sensors work in technology.

Parts of the books included.



Screen Lessons

Click on the link to learn more about the types of sound waves :

https://www.youtube.com/watch?v=QhGLt5uYg_U

Figure 12: Types of Sound Waves.

Dig Deeper

You just explored how sound waves can change depending on their frequency—and how some sounds are too high or too low for us to hear.

Now let's see how animals use sound in amazing ways!

Read this article from National Geographic:

https://www.nationalgeographic.com/animals/article/echolocation-is-nature-built-in-sonar-here-is-how-it-works

Screen Lessons

Objectives:

- Understand the differences between infrasound, audible sound, and ultrasonic waves.
- Recognize how sound frequency relates to what humans and animals can hear.

- Build foundational knowledge for understanding how ultrasonic sensors work.

Video Link: https://www.youtube.com/watch?v=QhGLt5uYg_U

Possible Strategies for Implementation:

1. **Pre-Watching: Set a Purpose.**
   - Teacher Prompts:
     - "Before we watch, think about this: Why can elephants hear sounds that we can't?"
     - "What do you already know about sound waves?"
   - Student Task:
     - Write down one question you have about sound waves before the video begins.
2. **During Watching: Active Engagement.**
   - Teacher Prompts:
     - Pause at the graph showing frequency and ask students to read the axis and explain what they notice.
     - Highlight terms like "frequency", "Hz", "ultrasonic", and "infrasound."
   - Student Task:
     - Fill in a quick sound wave chart:

Sound Type	Frequency Range	Who Hears It?
Infrasound	Below 20 Hz	Elephants, whales
Audible	20 Hz – 20,000 Hz	Humans
Ultrasound	Above 20,000 Hz	Bats, dolphins

3. **Post-Watching: Discussion and Reflection**
   - Class Discussion Questions:
     - "What surprised you the most about how animals use sound?"
     - "Why do you think scientists use ultrasonic waves in technology like sensors?"
   - Student Task:
     - Write a reflection or draw a quick sketch comparing human and animal hearing ranges.

Dig Deeper

Objective:

Understand how animals use sound—especially echolocation—as a survival tool, and how this connects to the concept of sound frequency (infrasound and ultrasound).

Article Link: https://www.nationalgeographic.com/animals/article/echolocation-is-nature-built-in-sonar-here-is-how-it-works

Possible Strategies for Implementation:

1. **Pre-Reading: Activating Prior Knowledge & Setting Purpose.**
   - Teacher Prompts:
     - o “What do you know about how animals ‘see’ or sense things without using their eyes?”
     - o “Have you ever heard of echolocation? Who might use it?”
   - Student Task:
     - o Predict how echolocation might work. Write or draw a short idea: "I think echolocation means..."
2. **During Reading: Focused Information Gathering.**
   - Teacher Instructions:
     - o Guide students to highlight or note key terms such as echo, frequency, ultrasound, sonar, and navigation.
     - o Pause periodically to ask: “What did the article just say about how animals use sound?”
   - Student Task:
     - o Use a simple note sheet with prompts like:

   1. Which animals use echolocation?
   2. Why do they use it?
   3. What part of the article surprised you?
3. **Post-Reading: Consolidating Learning & Reflection.**
   - Teacher Prompts:
     - o “Why might echolocation be better than eyesight in some situations?”
     - o “How can this connect to the way we use technology (like sensors or sonar)?”
   - Student Task:
     - o Complete one of the following:

   1. Draw a comparison between how bats and dolphins use echolocation.
   2. Write a short paragraph explaining why echolocation is important for survival.
   3. Create a “Did You Know?” fact card with one amazing echolocation fact and a visual.

Elaborate

Teacher Role:

- Guide students in applying scientific formulas (distance, time, speed) through real-world animal echolocation scenarios.
- Support student understanding of sound wave reflection and transmission in air and water.
- Facilitate the transition from mathematical problem-solving to coding a real ultrasonic distance sensor using Arduino.
- Prompt students to observe, test, and refine their simulations, highlighting how sensors mimic biological echolocation systems (like bats and dolphins).
- Scaffold code interpretation and Arduino simulation tasks, especially where students need to troubleshoot missing lines or logic conditions.

Expected From Students:

- Apply the formula: Distance = (Speed x Time)/2 to solve echolocation-based word problems involving animals and submarines.
- Accurately identify missing variables or lines in the Arduino code and explain each part's purpose.
- Build and simulate a distance-measuring circuit using an ultrasonic sensor in Tinkercad or on real hardware.
- Interpret the Serial Monitor outputs and adjust the code to explore how object distance affects the response.
- Complete a functional code and modify thresholds to trigger sound responses using a buzzer (e.g., buzzer turns on when object is closer than 10 cm).
- Reflect on the connection between real animal echolocation and how we simulate it in technology.

Parts of the books included.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Assessment

Apprentice

Calculate using the formula:

Distance = (Speed x Time) / 2

Figure 13: Measuring Water Depth Using SONAR

Figure 14: Echolocation in Nature and Technology.

A bat makes a squeaky sound, and the echo comes back after 2 seconds. If sound travels 10 meters every second, how far away was the object that reflected the sound?

A dolphin is using echolocation to find a yummy fish. The fish is 30 meters away. Sound travels 1,500 meters per second in water.

How long does it take for the sound to reach the fish and return as an echo?

A submarine sends out a sonar signal to find a sunken ship. The ship is 1,000 meters away, and the echo comes back after 1.33 seconds.

Can you figure out the speed of sound in water based on this information?

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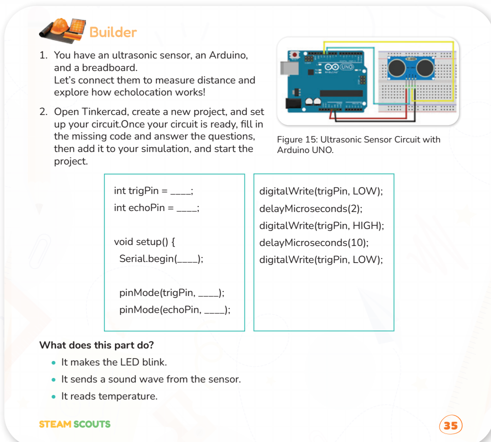
Builder

1. You have an ultrasonic sensor, an Arduino, and a breadboard.
   Let's connect them to measure distance and explore how echolocation works!
2. Open Tinkercad, create a new project, and set up your circuit.Once your circuit is ready, fill in the missing code and answer the questions, then add it to your simulation, and start the project.

Figure 15: Ultrasonic Sensor Circuit with Arduino UNO.

```
int trigPin = ____;
int echoPin = ____;

void setup() {
  Serial.begin(____);

  pinMode(trigPin, ____);
  pinMode(echoPin, ____);
```

```
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
```

What does this part do?

- It makes the LED blink.
- It sends a sound wave from the sensor.
- It reads temperature.

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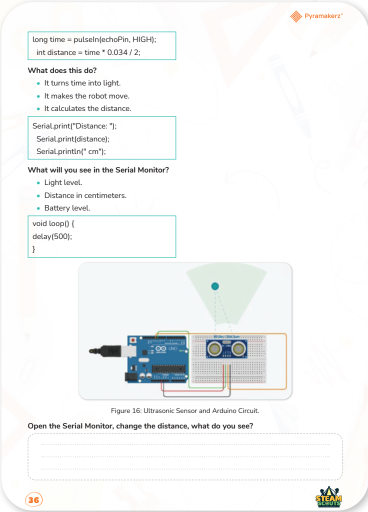

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```
long time = pulseIn(echoPin, HIGH);
 int distance = time * 0.034 / 2;
```

What does this do?

- It turns time into light.
- It makes the robot move.
- It calculates the distance.

```
Serial.print("Distance: ");
 Serial.print(distance);
 Serial.println(" cm");
```

What will you see in the Serial Monitor?

- Light level.
- Distance in centimeters.
- Battery level.

```
void loop() {
delay(500);
}
```

Figure 16: Ultrasonic Sensor and Arduino Circuit.

Open the Serial Monitor, change the distance, what do you see?

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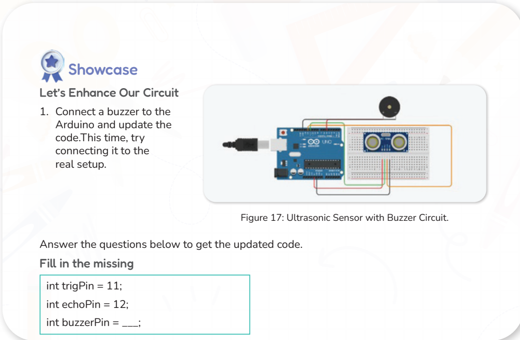


Showcase

Let's Enhance Our Circuit

1. Connect a buzzer to the Arduino and update the code.This time, try connecting it to the real setup.

Figure 17: Ultrasonic Sensor with Buzzer Circuit.

Answer the questions below to get the updated code.

Fill in the missing

```
int trigPin = 11;
int echoPin = 12;
int buzzerPin = ___;
```

Apprentice

Objective:

- Apply the formula Distance = (Speed × Time) ÷ 2 to real-world echolocation problems.
- Understand how animals and machines (like bats, dolphins, and submarines) use reflected sound waves to determine distance.
- Strengthen understanding of how time, speed, and distance are connected.

Possible Answers:

1. **Bat Question:**
   - Time = 2 seconds
   - Speed = 10 m/s
   - Distance = (10 × 2) ÷ 2 = 10 meters.
   - (The object is 10 meters away because 2 seconds is for the full trip to the object and back.)
2. **Dolphin Question:**
   - Distance = 30 meters (one way) → 60 meters round trip
   - Speed = 1,500 m/s
   - Time = (60 ÷ 1,500) = 0.04 seconds
3. **Submarine Question:**
   - Distance = 1,000 meters
   - Total time = 1.33 seconds
   - Since it's round-trip, one way = 1,000 meters
   - Speed = (2 × 1,000) ÷ 1.33 = ~1,503.76 m/s
   - (This confirms the real speed of sound in water ≈ 1,500 m/s.)

Possible Strategies for Implementation:

1. **Kagan Structure – Rally Coach:**
   - Pair students. One student solves while the other coaches, then they switch. This ensures both practice the formula and explain their reasoning.
2. **Visible Thinking – "Claim–Support–Question"**
   - After solving each scenario, students state:
     - Claim: The answer they calculated.
     - Support: The steps and logic used to solve.
     - Question: What would happen if the speed or time changed?
3. **Real-World Connection & Comparison Discussion:**
   - Ask: Why do dolphins need such fast sound speeds? Why don't humans hear echoes like bats?
   - Show how scientists use the same principles in sonar, radar, and even in medical imaging (like ultrasounds).
4. **Graphic Organizer – Table Completion:**
   - Prepare a table with columns: Object/Animal – Speed – Time – Distance – Calculated? (Y/N)
   - Students fill in values and see patterns.
5. **Error Checking Challenge:**
   - Provide a "wrong" example (e.g., forgetting to divide by 2) and let students explain what's incorrect and why.

Builder

Objective:

Students will connect an ultrasonic sensor to an Arduino using Tinkercad, write and complete code to calculate distance using sound waves, and interpret serial monitor output to understand how echolocation works.

Answers:

```
int trigPin = 9;
int echoPin = 10;
void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}
```

```
void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long time = pulseIn(echoPin, HIGH);
  int distance = time * 0.034 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  delay(500);
}
```

Question Responses:

1. What does this part do?

```
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
```

B. It sends a sound wave from the sensor.

2. What does this do?

```
cpp
CopyEdit
long time = pulseIn(echoPin, HIGH);
int distance = time * 0.034 / 2;
```

C. It calculates the distance.

3. What will you see in the Serial Monitor?

B. Distance in centimeters

Open the Serial Monitor, change the distance, what do you see?

Answer:

When I move the object closer, the number goes down. When I move it farther away, the number goes up. The sensor is measuring the distance in centimeters using sound waves!

Activity Instructions:

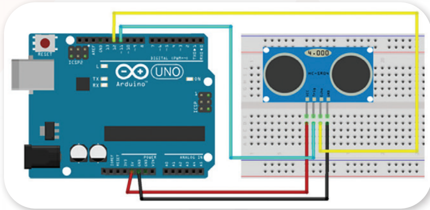


Figure 15: Ultrasonic Sensor Circuit with Arduino UNO.

1. Connect the Circuit:
   - Use the ultrasonic sensor image as a guide.
   - Connect the Trig and Echo pins to Arduino digital pins (e.g., Trig → D9, Echo → D10).
   - Connect VCC to 5V and GND to GND.
2. Open Tinkercad Circuits.
   - Create a new Arduino project.
   - Drag and drop an ultrasonic sensor and Arduino board.
   - Build the circuit and name your file.
3. Complete the Code:
   - Fill in the missing pins:

```
int trigPin = 9;
int echoPin = 10;
```

   - Fill in the rest of the code blocks as shown in the student book.
   - Upload and start the simulation.
4. Monitor Output:
   - Open the Serial Monitor in Tinkercad.
   - Move the object closer/farther from the sensor.
   - Observe and record how the distance value changes.
5. Answer Checkpoint Questions:
   - Select correct answers from the multiple-choice questions.
   - Write a reflection on what you observed when changing distances.

Possible strategies to implement:

1. Think–Pair–Share:

- Have students predict what the sensor will display when an object moves before testing it. Then, compare actual output after running the simulation.

2. See–Think–Wonder Routine:

- Ask:
  - o What do you see in the code?
  - o What do you think each part does?
  - o What do you wonder about distance and timing?

3. **Inquiry Station Rotation:**
   - Set up small groups to each build the circuit, simulate, observe, and record different object distances. Rotate and share findings.
4. **Visual Modeling:**
   - Use visual aids and diagrams to demonstrate the trig/echo signal. Model how sound bounces and is timed.
5. **UDL Tip:**
   - Allow students to either:
     - Watch a pre-recorded simulation walkthrough,
     - Follow text steps with images, or
     - Use a physical Arduino (if available) for tactile learners.

Evaluate

Teacher Role:

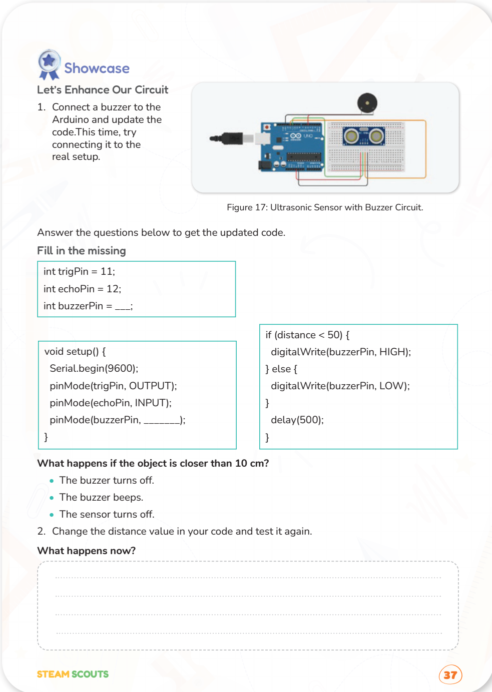
- Guide students through testing the buzzer-enhanced circuit and make sure connections follow the diagram.
- Prompt students to predict what will happen when the object gets close to the sensor, before they run the code.
- Ask guiding questions:
  - "Why does the buzzer turn on or off?"
  - "What value controls the buzzer behavior?"
  - "How does changing the number in the code affect the result?"
- Encourage students to observe, describe, and reflect on how the code and circuit respond to distance changes.
- Support students as they experiment with modifying the distance value in the if-condition to observe behavior shifts.
- Facilitate a short group discussion or pair-share reflection after testing.

Expected From Students:

- Actively participate in completing the missing parts of the code and uploading it to the Arduino.
- Test their buzzer circuit and observe how it behaves when an object moves closer than the set distance.
- Answer the question: What happens if the object is closer than 10 cm?
- Change the distance value (e.g., from 50 to 20 or 80), upload again, and record observations.

- Discuss their observations using vocabulary such as “condition,” “distance,” “trigger,” and “output.”
- Reflect on how small code changes impact real-world outputs.

Parts of the books included.


Showcase

Let’s Enhance Our Circuit

1. Connect a buzzer to the Arduino and update the code.This time, try connecting it to the real setup.

Figure 17: Ultrasonic Sensor with Buzzer Circuit.

Answer the questions below to get the updated code.

Fill in the missing

```
int trigPin = 11;
int echoPin = 12;
int buzzerPin = ___;
```

```
void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, _______);
}
```

```
if (distance < 50) {
  digitalWrite(buzzerPin, HIGH);
} else {
  digitalWrite(buzzerPin, LOW);
}
  delay(500);
}
```

What happens if the object is closer than 10 cm?

- The buzzer turns off.
- The buzzer beeps.
- The sensor turns off.

2. Change the distance value in your code and test it again.

What happens now?

STEAM SCOUTS 37

Showcase

Objective:

Students will apply their understanding of input/output pins and conditional logic by adding a buzzer to the Arduino circuit. They will observe how changes in code conditions affect hardware behavior and reflect on system feedback and distance thresholds.

Possible Answers:

1. Fill in the missing:

```
cpp
CopyEdit
int buzzerPin = 7; // or the pin they used in their circuit
```

2. pinMode setup:

```
cpp
CopyEdit
pinMode(buzzerPin, OUTPUT);
```

3. What happens if the object is closer than 10 cm?

B. The buzzer beeps

4. What happens when the distance value is changed?

Sample Answer:

- When I changed the value to 80, the buzzer turned on even when the object was farther away. When I changed it to 20, it only turned on when the object got very close.

Activity Instructions:

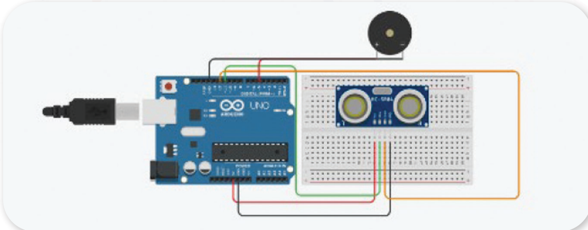


Figure 17: Ultrasonic Sensor with Buzzer Circuit.

1. Review the Existing Circuit:

- Begin by revisiting the ultrasonic sensor setup with the class. Ask students what each component does. Then introduce the buzzer as a new output device.

2. Guide the Buzzer Connection:

- Show the breadboard and Arduino diagram provided.
- Walk students through identifying an available digital pin (e.g., pin 7).
- Support students in connecting the buzzer's positive lead to the chosen digital pin and negative lead to GND.

3. Code Completion Support:

- Display the code with missing parts on the board or projector.
- Encourage students to fill in:

```
cpp
CopyEdit
int buzzerPin = 7;
pinMode(buzzerPin, OUTPUT);
```

- Clarify the purpose of these lines and how the buzzer is being added to the system.

4. Testing the Circuit:

- Ask students to upload the full code and place an object at different distances from the sensor.
- Instruct them to observe the buzzer's response when the object moves closer than 50 cm.

5. Prompt Reflection Question:

- Ask: "What happens if the object is closer than 10 cm?"
- Provide answer choices and discuss why the correct answer is "The buzzer beeps."

6. Code Modification & Rerun:

- Prompt students to experiment by changing the distance threshold in the code (e.g., from 50 to 20 or 80).
- Let them upload and test again to compare how the buzzer's behavior changes.

7. Support Student Discussion & Documentation:

- Have students write their observations using sentence starters:
    - "When I changed the value to __, the buzzer started/stopped at __ cm."
- Encourage pair discussions or small group sharing to reinforce reasoning.

Possible Strategies to Implement:

1. Think-Pair-Share:

- Before uploading the code, ask students to predict what will happen and discuss in pairs. After testing, they revisit their predictions and refine their understanding.

2. See-Think-Wonder:

- Use this Visible Thinking routine during testing:
    - What do you see happening in the circuit?
    - What do you think the code is doing?
    - What do you wonder about the buzzer and sensor behavior?

3. Code → Test → Reflect Loop:

- Encourage students to iteratively update the code, test it, and document changes in behavior. This models real engineering debugging processes.

4. Gallery Walk (Optional Extension):

- Students walk around and observe how their peers set different threshold values. They compare circuit behavior and code snippets and discuss.

Rubric: Arduino Buzzer Circuit and Conditional Logic.

Criteria	Needs Improvement (1 Point)	Satisfactory (2 Points)	Very Good (3 Points)	Excellent (4 Point)
Circuit Completion (10 pts).	Buzzer is not connected or wired incorrectly.	Buzzer is connected with help; minor issues in wiring.	Buzzer is connected correctly with minimal help.	Buzzer is correctly wired and tested independently.
Code Accuracy (10 pts).	Missing or incorrect buzzerPin declaration or pinMode.	Code mostly works; includes both lines but may have syntax or logic errors.	Complete code with accurate pin declaration and logic; minor support needed.	Fully accurate code including buzzerPin, pinMode, and working if-condition.

Testing & Debugging (10 pts).	Little or no testing; issues not identified.	Basic testing done; needs support to notice or fix errors.	Tests circuit and reflects on buzzer behavior; minor debugging help needed.	Independently tests, identifies issues, modifies code, and explains fixes clearly.
Observation & Reflection (10 pts).	Limited or no observations recorded; unclear explanation.	Some observations made; responses need more detail or reasoning.	Observations are recorded with reasoning (e.g., how distance affects buzzer).	Thoughtful, specific observations with clear explanation of cause-effect between code and behavior.
Application of Logic (10 pts).	Does not demonstrate understanding of how distance affects output behavior.	Shows basic understanding but struggles to apply it in code.	Demonstrates conditional logic by adjusting threshold and explaining outcomes.	Applies conditional logic skillfully and explains how and why system behavior changes with code edits.

Don't forget to go back to the EDP page and add these to creating part.

Create

I connected the ultrasonic sensor and buzzer to the Arduino using jumper wires and a breadboard. I made sure each component was placed in the right position so the circuit could work properly. Then, I added a new line in the code to define the buzzer pin and used an if-condition to control when the buzzer would turn on or off based on the distance. After uploading the code, I tested how the system reacted when an object came close to the sensor. The buzzer beeped when the object was within the set range. If something didn't work at first, I checked my wiring, fixed the code, and tested again until it worked correctly.

Now I Can

Objective:

Students will consolidate their learning by reflecting on key science and engineering concepts explored in the lesson.

Activity Instructions:

1. Set the Scene for Reflection:

- After students have completed the circuit testing and coding tasks, bring the class together to revisit the goals of the lesson. Display the “Now I Can...” list or read each statement aloud.

2. Prompt Self-Assessment:

- Ask students to reread each “Now I Can” statement silently. Then guide them to place a ✓ next to each one they feel confident about. For any they’re unsure of, encourage them to write a question or an example that still confuses them.

3. Visible Thinking Strategy – “I Used to Think... Now I Think...”

- Have students complete this sentence in their journals or on sticky notes:
  - “I used to think ultrasonic sensors were just like regular sound devices, now I think...”
  - “I used to think distance was measured by rulers, now I think...”

4. Optional Group Discussion or Sharing Circle:

- Invite volunteers to share their thoughts on which “Now I Can” statement was most interesting or surprising to them, and why.

5. Quick Application Challenge (Optional Extension):

Ask: “Where else could we use an ultrasonic sensor in the real world?” Have students brainstorm real-life uses, linking their learning to smart devices, parking systems, or robotics.

Relation to Standards:

1. NGSS (Next Generation Science Standards):

- **PS4.A:** Wave Properties.
  - Students explore the properties of sound waves (frequency, speed) and how they travel through different materials, relating to how ultrasonic sensors function.
- **ETS1.A:** Defining and Delimiting Engineering Problems.
  - Through hands-on prototyping and code modification, students define problems and test potential solutions (e.g., alerting users using a buzzer when an object is too close).
- **ETS1.B:** Developing Possible Solutions.
  - Students iteratively build and refine a working sensor circuit using Arduino to simulate real-world systems.

2. ISTE Standards for Students:

- **1.4 Innovative Designer**
  - Students use Arduino and coding to solve problems and create a real-world application that responds to sensor input.
- **1.5 Computational Thinker**
  - They apply logic (if-else conditions), estimation, and debugging to control the buzzer based on sensor data, modeling algorithmic thinking.
- **1.1 Empowered Learner**
  - Students reflect on their learning using the "Now I Can" checklist, showing ownership of their progress and understanding.

3. Common Core Math Standards:

- **CCSS.MATH.CONTENT.5.MD.A.1**
  - Students convert and use measurements (e.g., time, distance, speed of sound) to estimate and calculate real-world values from sensor data.

4. UNESCO ICT Competency Framework:

- **Technology Literacy:** Apply digital tools to real-world problems.
  - Students build and adapt an Arduino-powered sensing system to simulate real environmental detection systems.

5. Sustainable Development Goals (SDGs):

- **SDG 9:** Industry, Innovation and Infrastructure.
  - Students explore how sensor technologies are used in engineering solutions and smart systems.
- **SDG 4.7**: Education for Sustainable Development and Global Citizenship.
  - By reflecting on real-life uses of ultrasonic sensors (e.g., for blind navigation, safety, robotics), students connect their learning to global technology challenges and inclusive design.